

FCU3 Fabrication Document

Layer Stack Legend

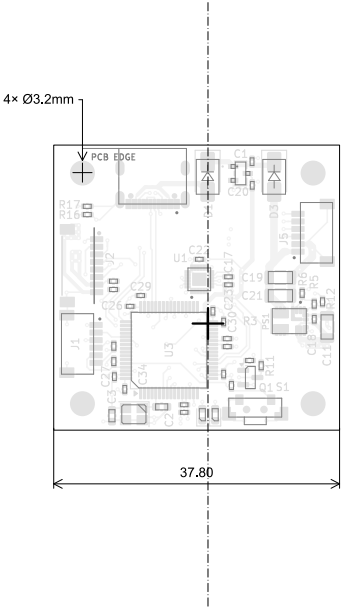
	Material	Layer	Thickness	Dielectric	Type	Gerber
	F,Paste				Paste Mask	
	F,Silkscreen			Direct Printing	Legend	GBR
	F,Mask		0,02mm	Solder Resist	Solder Mask	GBR
	Copper	L1 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
	Prepreg		0,1mm	FR4_7628	Dielectric	
	Copper	L2 (GND)	0,035mm (1,00oz)		Plane	GBR
	Core		1,21mm	FR4	Dielectric	
	Copper	L3 (Sig, PWR)	0,035mm (1,00oz)		Plane	GBR
	Prepreg		0,1mm	FR4_7628	Dielectric	
	Copper	L4 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
	B,Mask		0,02mm	Solder Resist	Solder Mask	GBR
	B,Silkscreen			Direct Printing	Legend	GBR
	B,Paste				Paste Mask	

Total thickness: 1,66mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0,2032	0,28	L2

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.

B. Tg 170 C OR EQUIVALENT.

C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY KALDSTRAND ELECTRONICS.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE

37.800 × 37.700 mm

BOARD THICKNESS

1.660 mm

TRACE WIDTH

0.200 mm

TRACE TO TRACE

0.200 mm

MIN. HOLE (PTH)

0.200 mm

MIN. HOLE (NPTH)

0.650 mm

ANNULAR RING

0.050 mm

COPPER TO HOLE

0.254 mm

COPPER TO EDGE

0.250 mm

HOLE TO HOLE

0.254 mm

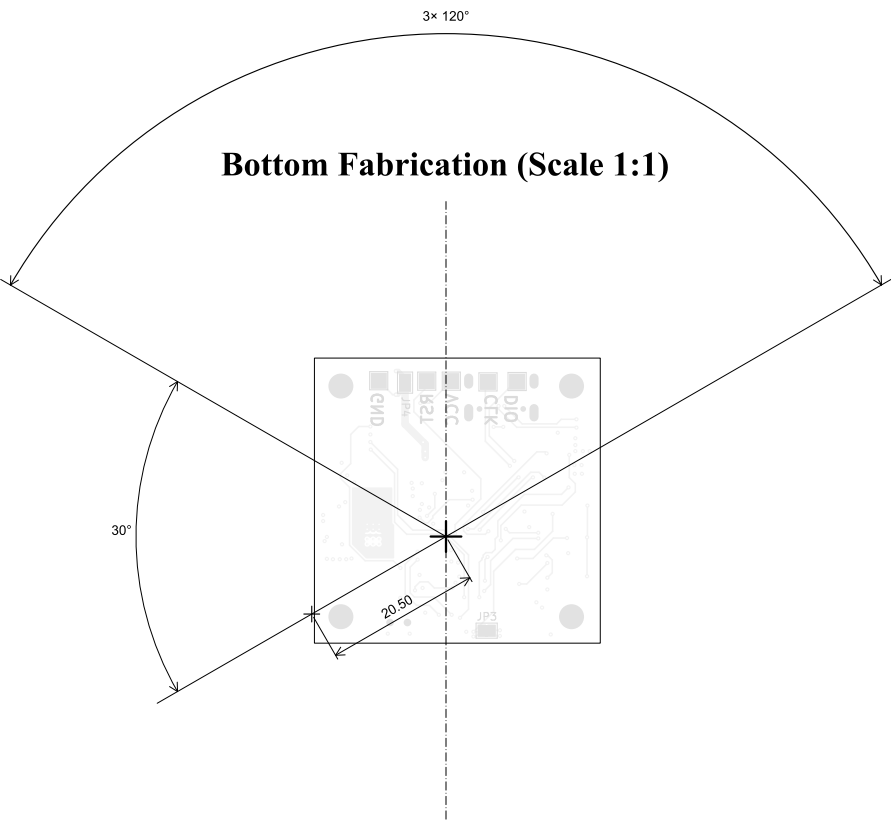
11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 0c4f9ce
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Top Fabrication (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 1 of 10

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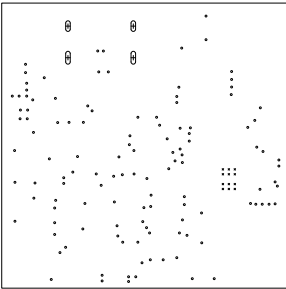
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		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 2 of 10

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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	12	0,20mm (7,87mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
○	121	0,25mm (9,84mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Via
+	4	0,60mm (23,62mils)	PTH	Slot	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
Total 137						

Drill Drawing L1 - L4 (Scale 1:1)



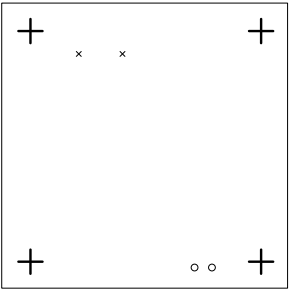
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		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 3 of 10

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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	2	0,85mm (25,59mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
O	2	0,90mm (35,43mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
+	4	3,20mm (125,98mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
Total 8						

Drill Drawing L1 - L4 (Scale 1:1)



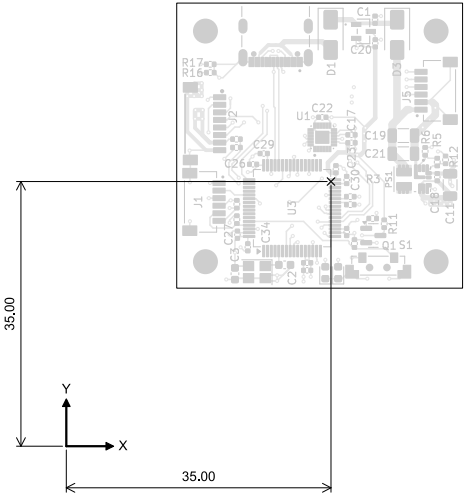
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	Sheet Title: Drill Drawing (L1 - L4)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
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FCU3 Fabrication Document

Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
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Ref.	Net	X [mm]	Y [mm]
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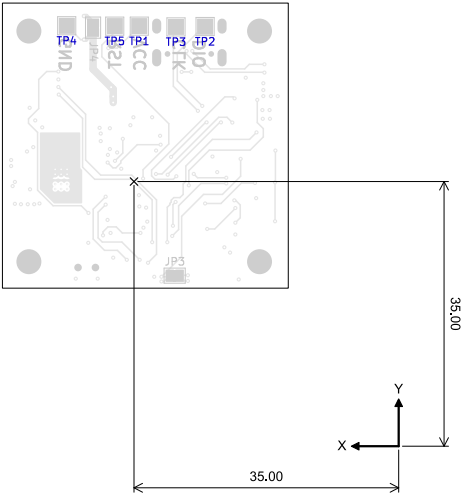
All dimensions are in millimeters unless otherwise specified.

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	Sheet Title: Top Test Points (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 5 of 10

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Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP1	+3V3	19.70	-3.05
TP2	SWDIO	11.00	-3.10
TP3	SWCLK	14.85	-3.15
TP4	DGND	29.30	-2.99
TP5	Net-(U3-NRST)	22.95	-3.05

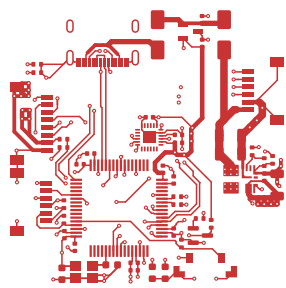


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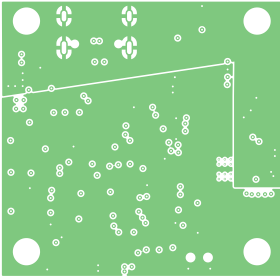
L1 (Sig, PWR) (Scale 1:1)



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		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: L1 (Sig, PWR) (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
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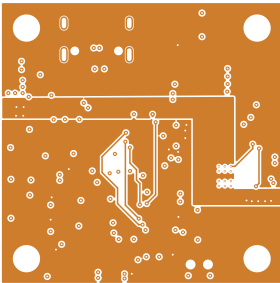
L2 (GND) (Scale 1:1)



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		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: L2 (GND) (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
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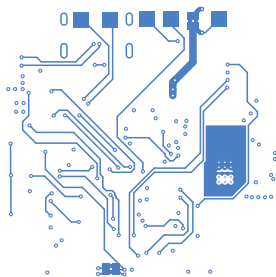
L3 (Sig, PWR) (Scale 1:1)



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		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: L3 (Sig, PWR) (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
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L4 (Sig, PWR) (Scale 1:1)



	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 0c4f9ce
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: L4 (Sig, PWR) (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
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